

Linksys WRT300N

Feature-rich, future-proof

The WRT300N is a stylish, futuristic-looking router that supports the yet-to-be-announced 802.11n standard (Pre-n as of now). Its dark grey, black and smoky blue body will turn eyes, as will the radar-looking antenna booster.

The WRT300N brims over with features that can be accessed from its intuitive firmware menu. The WRT300N also supports MIMO, which helps maximise signal coverage while reducing occurrences of dead spots within its effective range.

We expected a lot from this model, it being the only Pre-n device that made it to this test. As you'll see in our table, it brought up some solid scores, but nothing worth cheering about. It was well and truly beaten by some of the other routers—in particular the ASUS WL-530G and the Netgear DG834G.

The WRT300N remains a good choice for the home segment because of its plethora of features and the fact that it's future-proof.



Linksys WRT54G

Performance you don't pay a wad for!

The WRT54G is a simple, no-nonsense looking router that sports twin antennas. The firmware is as intuitive as that on the WRT300N. The minus point is immediately evident—there's no logout button; you simply have to close your browser window after saving settings (in case you've changed any). We weren't able to check from the interface whether our two laptops were connected. For that, we had to filter the MAC address list to show the clients connected. A couple of things missing in the interface; not of great

use to home users, but a SoHo would miss them. Incidentally, the WRT54G loses out on Wireless Bridging as a feature.

While it outpaces the WRT300N in the Zone 1 tests, it lags behind in Zone 2. This shows its signal quality is slightly inferior, because signal strength of the models in Zone 2 was identical. Where this one scores is in pricing; it costs just Rs 3,634, and you'll get why the Digit Best Buy Silver award goes to the WRT54G.



How We Tested

Testing Wireless devices involves gauging performance on mainly two fronts—transfer rates (speed) and effective range.

Our Wi-Fi test was done simulating all possible real-world conditions, for which we created the test “zones” we've described below. Please note that distances aren't important for benchmarking any sort of Wi-Fi network; it's the obstructions between the Access Point (AP) and the signal recipient that ultimately matters. Metallic objects, wooden cabinets, concrete walls, and general furniture all cause what is called attenuation of wireless signals. The signal strength is reduced, which causes loss of data integrity and therefore slows down data throughput. With this in mind, we designed each of these zones to actually take each of these wireless devices to their performance limits.

Zone 1: The laptop was placed within 15 feet of the AP. There was a wooden partition between the AP and the client, though they were essentially in the same room. Signal attenuation was at a minimum.

Zone 2: We placed the laptop within 25 feet of the router. This time we ensured there were a good deal of signal-obstructing objects. There was a concrete wall (with inbuilt cupboards) in between the Wi-Fi devices to further scramble reception. This was in addition to a couple of wooden partitions.

Zone 3: Our test laptop was placed approximately 30 feet away from the router. Here we actually moved the laptop to a higher floor. We also ensured the laptop wasn't exactly above the router placed a level below (this would be too easy). This was the most strenuous test for the routers, as multiple obstructions create ample cause for attenuation and hence signal loss.

We created test files to be copied from the PC to the laptop wirelessly. In each Zone we used two different files. One was a

single RAR archive, while the other was a set of small files of various types to simulate every possible data transfer scenario. The file sizes were kept the same in the same Zone, but sizes across Zones varied—100 MB for Zone 1, 50 MB for Zone 2, and 25 MB for Zone 3. The speeds were recorded in Megabytes per second (MBps). Please note that most manufacturers quote figures in Megabits per second (Mbps). Keep in mind that 1 byte equals 8 bits.

Next we streamed an HD video file encoded at 1280 x 720 (720p). This clip was on our test PC's hard disk, and was streamed to first one, and then two, laptops. We awarded points based on the entire viewing experience. In case any jerking of frames was really noticeable, points were deducted accordingly. A score in excess of

Test Rig Specifications

	PC	Laptop 1	Laptop 2
Processor	AMD x2 4800+	Pentium M 1.73 GHz	Pentium M 1.2 GHz
Motherboard	AMD x2 4800+	Intel 915GMS	Intel 915GMS
Memory	512 MB x 2 Corsair XMS DDR 400	512 MB DDR2 533	512 MB DDR2 533
Graphics card	XFX Geforce 7900GTX	Intel Integrated 915GM	Intel Integrated 915GM
HDD	Hitachi T7K 250 SATA 2.0	Toshiba 80 GB 4200 rpm	Fujitsu 60 GB 4200 rpm
Wi-Fi Solution	NA	Intel Integrated	Intel Integrated

7.5/10 means the device is capable of providing a good movie-watching experience.

Finally, we used Wireless Mon version 2.0 as a benchmarking tool to measure signal strength in each of the Zones.

In our signal strength at 20 metres test, we ensured there were two concrete walls between the AP and the test laptop. Wireless Mon was our benchmark of choice here as well.